

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Week No.: 4

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1. TITLE

Develop Dynamic Webpage Using JavaScript

2. AIM

To build an interactive dynamic webpage with real-time clock updating and dark theme toggling.

3. OBJECTIVE

The objectives of this experiment are:

- Implement DOM event listeners.
- Utilize setInterval for live time updates.

4. THEORY

JavaScript event listeners and DOM manipulation enable real-time UI interactions.

5. ALGORITHM / PROCEDURE

1. Create index.html.
2. Write updateClock() using setInterval.
3. Add theme button click handler.
4. Test in browser.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>

<head>
  <title>Dynamic Web Application</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: Arial, sans-serif;
      background: #f4f7fb;
```

```
padding: 20px;
color: #222;
}

header {
  background: #1e293b;
  color: white;
  padding: 15px;
  text-align: center;
  border-radius: 8px;
}

.box {
  background: white;
  padding: 20px;
  margin: 15px 0;
  border-radius: 8px;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
}

input {
  padding: 8px;
  width: 60%;
  margin-right: 8px;
}

button {
  padding: 8px 14px;
  background: #2563eb;
  color: white;
  border: none;
  border-radius: 4px;
  cursor: pointer;
}

button:hover {
  background: #1d4ed8;
}

#clock {
  font-size: 26px;
  font-weight: bold;
  margin-top: 10px;
  color: #2563eb;
}

li {
  margin: 6px 0;
  padding: 6px;
  background: #f1f5f9;
  border-radius: 4px;
}
</style>
</head>

<body>

<header>
```

```
<h1>Dynamic Web Application</h1>
<p>Interactive DOM Operations</p>
</header>

<div class="box">
  <h3>Student Greeting</h3>
  <input id="userName" placeholder="Enter your name">
  <button onclick="greetUser()">Greet</button>
  <p id="greetMsg" style="margin-top: 10px; font-weight: bold;"></p>
</div>

<div class="box">
  <h3>Digital Clock</h3>
  <div id="clock">00:00:00</div>
</div>

<div class="box">
  <h3>Task List</h3>
  <input id="taskInput" placeholder="Add new task">
  <button onclick="addNewTask()">Add Task</button>
  <ul id="taskContainer" style="margin-top: 10px; list-style: none;"></ul>
</div>

<div class="box">
  <h3>Theme Switcher</h3>
  <button onclick="setTheme('#f4f7fb', '#222')">Light</button>
  <button onclick="setTheme('#0f172a', '#f8fafc')">Dark</button>
  <button onclick="setTheme('#ecfdf5', '#064e3b')">Emerald</button>
</div>

<script>
  function greetUser() {
    const name = document.getElementById('userName').value;
    document.getElementById('greetMsg').textContent = name ? "Hello, " + name + "! Welcome to ANITS."
: "";
  }

  function showTime() {
    document.getElementById('clock').textContent = new Date().toLocaleTimeString();
  }
  setInterval(showTime, 1000);
  showTime();

  function addNewTask() {
    const val = document.getElementById('taskInput').value.trim();
    if (val) {
      const li = document.createElement('li');
      li.textContent = val;
      document.getElementById('taskContainer').appendChild(li);
      document.getElementById('taskInput').value = '';
    }
  }

  function setTheme(bg, color) {
    document.body.style.background = bg;
    document.body.style.color = color;
  }
</script>
```

</body>

</html>

7. CODE EXPLANATION

Code / Syntax	Explanation
<code>setInterval(updateClock, 1000)</code>	Updates clock every second
<code>classList.toggle('dark')</code>	Toggles dark theme CSS class

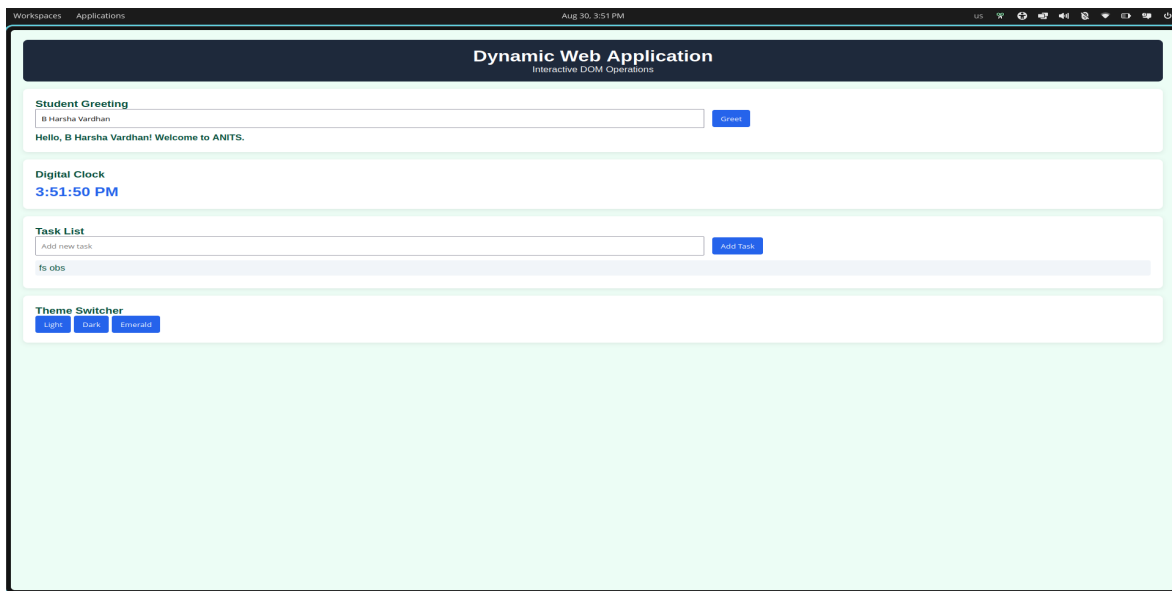
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Interactive Click	Theme toggles and clock updates	Theme toggles and clock updates	Pass

9. OUTPUT

Output 1: Execution Screenshot



10. RESULT

Thus, the experiment for Week 4 (Develop Dynamic Webpage Using JavaScript) was successfully executed and verified.